

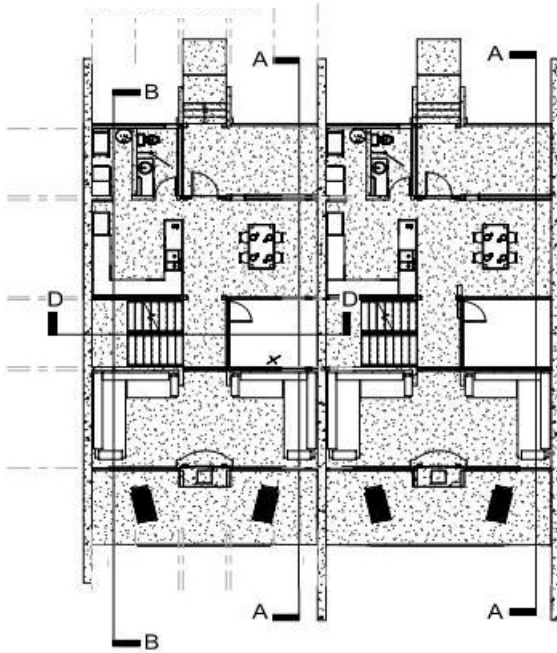
Salk Institute Center

ARC253, Architectural Studio Taught by Clyde Carpenter:

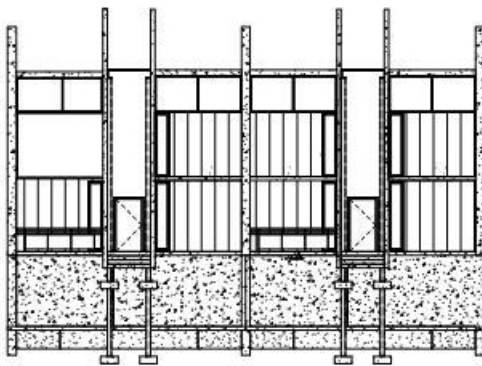
For this project, the students were asked to design houses for scientists working in the Salk institute center. The houses were required to share a common wall that separates the houses. Then, the students were asked to think of ways to aggregate 24 units of the house design. The students were also asked to provide a communal spaces for the residents and how the aggregation is designed for the site, the site is on top of a mountain over looking an ocean.



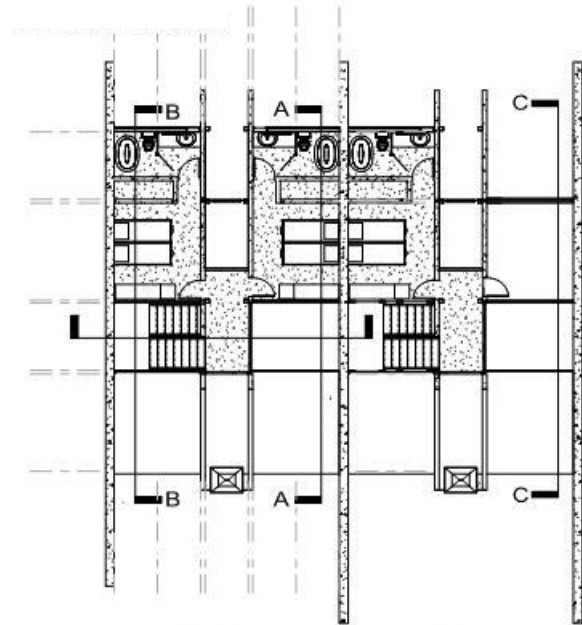
First Floor:



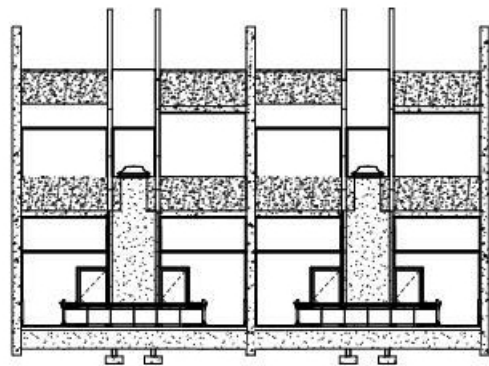
Front Elevation:



Second Floor:

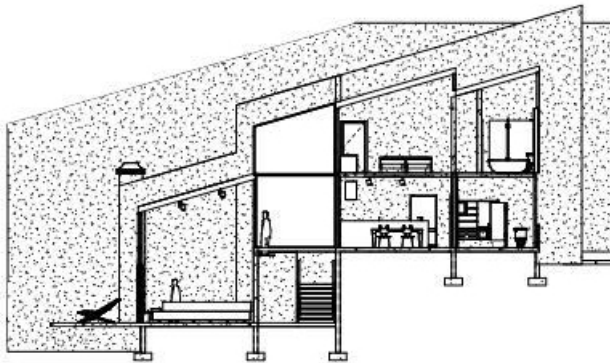


Back Elevation:

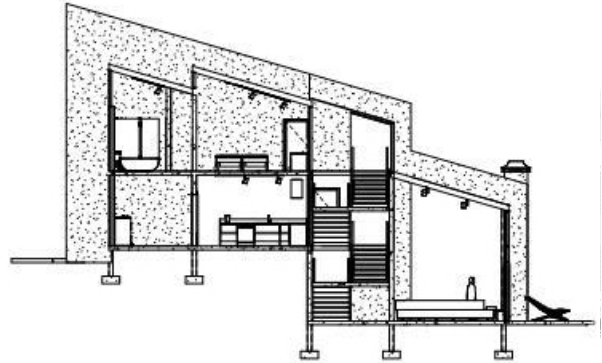


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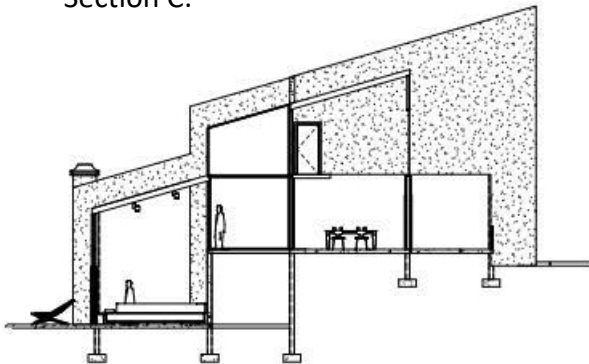
Section A:



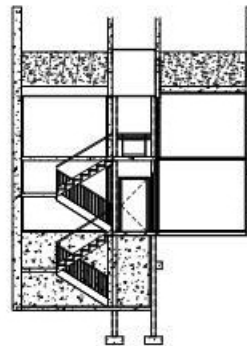
Section B:



Section C:



Section D:



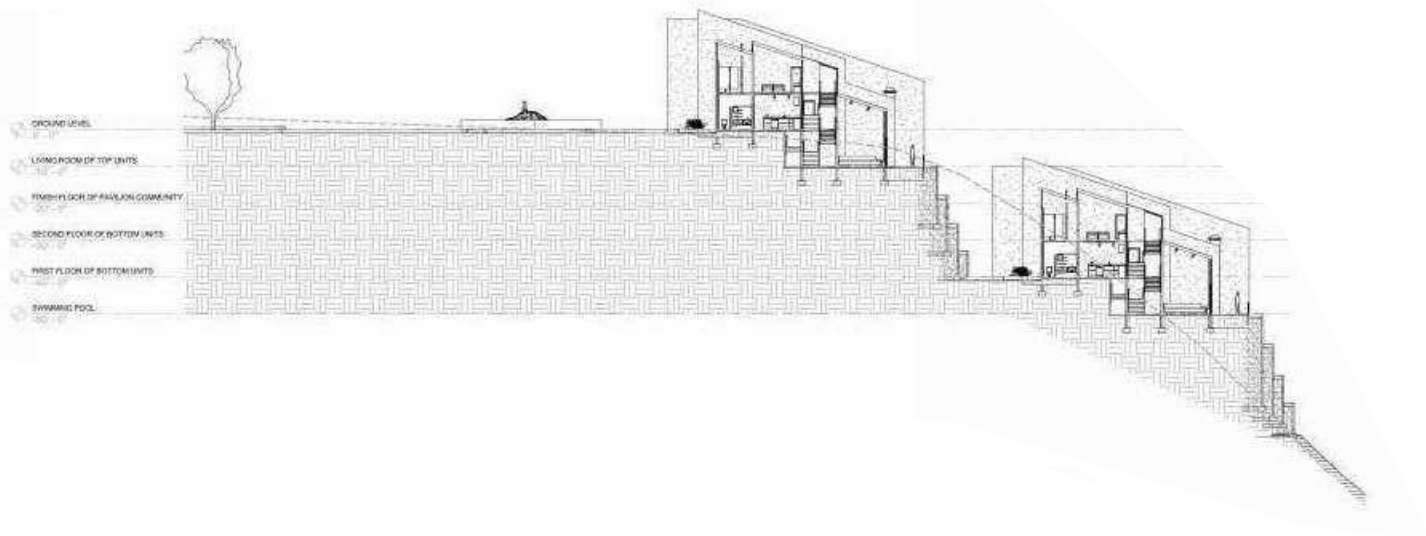


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Back Elevation:



Section 1:





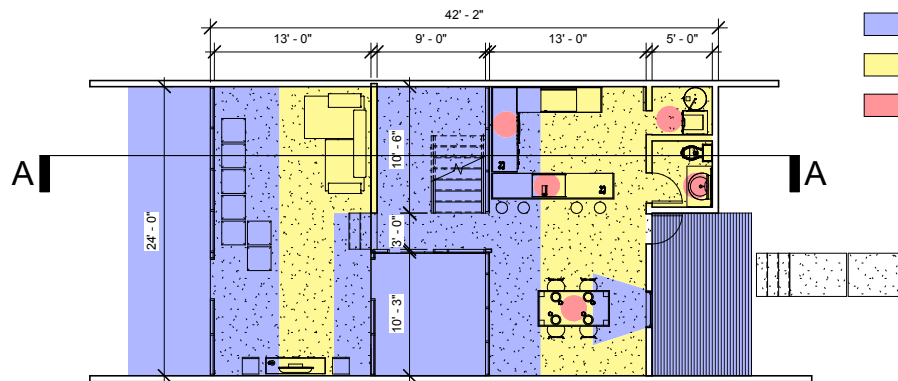
Top View:



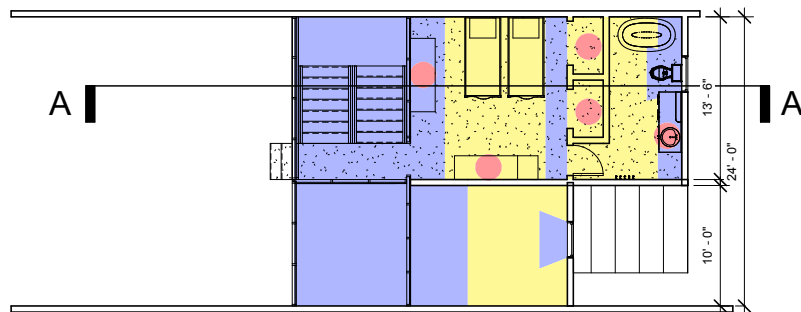
Integrated Lighting Diagrams

Key Plan:

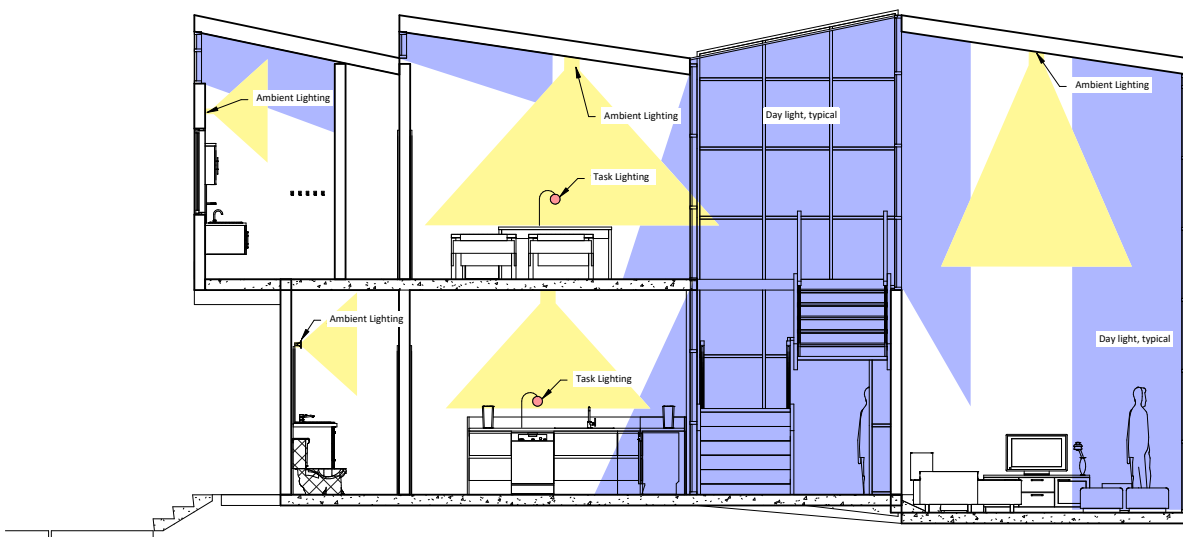
-  Illuminated by day lighting
-  Requires ambient lighting
-  Requires task lighting



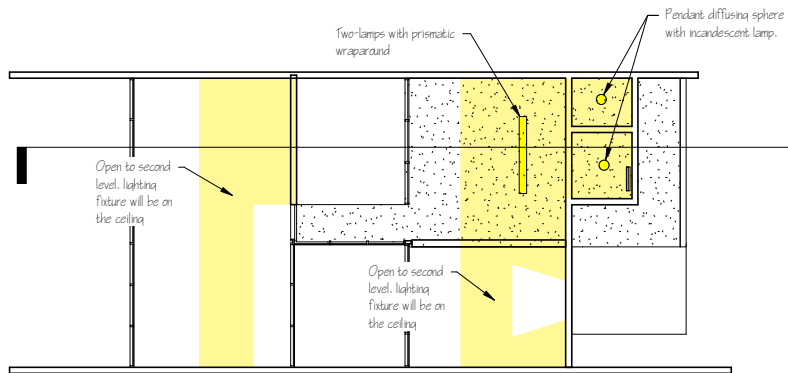
First Floor Plan



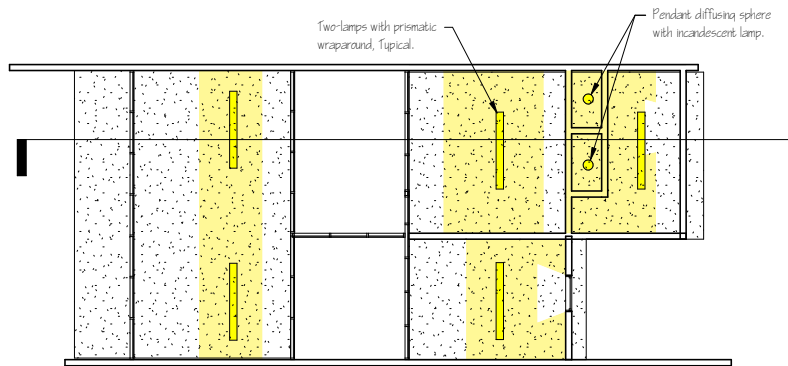
Second Floor Plan



Section A-A



Reflected Ceiling Plan of First Floor

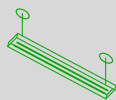


Reflected Ceiling Plan of Second Floor

Ambient Lighting:

Spaces	Sqft	No. of Luminaries	Lamps Per Luminaire	Type of Luminaire	Required F.C.
Bedrooms	140	1	2	fluorescent light - Two-lamps with prismatic wraparound	10-50
Living room	165	2	2	fluorescent light - Two-lamps with prismatic wraparound	10-50
Kitchen	110	1	2	fluorescent light - Two-lamps with prismatic wraparound	10-50
Dining	110	1	2	fluorescent light - Two-lamps with prismatic wraparound	10-40
Bathrooms	30	1	1	Pendant diffusing sphere with incandescent lamp.	10-30

Fluorescent lamps



incandescent lamp



Task Lighting:

- Task light is required for desks with 30-40 F.C.
- Task light is required for kitchen sink and stove with 10-30 F.C.

Task Light Fixture



Calculations:

$$\text{No. of fixtures} = \frac{\text{area (sqft)} \times \text{required F.C. level}}{\text{lamps} \times \text{initial lumens} \times \text{C.U.} \times \text{M.F.}}$$

$$\text{No. of fixtures} = \frac{(140 + 165 + 110 + 110 + 30) \times (25 + 25 + 25 + 20 + 15)}{2 \times 3200 \times 30 \times .70}$$

$$\text{No. of fixture} = 45$$

$$\text{F.C.} = \frac{\text{no. of luminaires} \times \text{no. of lamps per luminaires} \times \text{initial lumens per lamp} \times \text{coefficient of utilization} \times \text{maintenance factor}}{\text{area in square feet}}$$

$$\text{F.C.} = \frac{45 \times 2 \times 3200 \times 30 \times .70}{555 \text{ sqft}}$$

$$\text{F.C.} = 109$$